

Ganlin Chen, Ph.D.



Education	University of Michigan, Ann Arbor, USA	Sep. 2021 – Jun. 2026
	Doctor of Philosophy in Materials Science & Engineering Joint degree in Scientific Computing	
	University of Michigan, Ann Arbor, USA	Sep. 2019 – Apr. 2021
	Master of Science in Materials Science & Engineering	
	Soochow University, China	Sep. 2015 – Jun. 2019
	Bachelor of Engineering in New Energy Materials and Devices	
Teaching	Graduate Student Instructor, MSE 350: Structure of Materials, Fall term in 2024.	
Service	Session Chair, Hume-Rothery Symposium, 153rd TMS Annual Meeting & Exhibition, Florida, 2024.	
Awards	Silver Prize on Poster Session in MMRI, Ann Arbor, Michigan, 2026	
	Rackham Conference Travel Grant Award, 2025	
	Rackham Conference Travel Grant Award, 2024	
	Rackham Conference Travel Grant Award, 2023	
	Rackham Conference Travel Grant Award, 2022	
	Excellent Graduate Award of Soochow University, 2019	
Research	<i>Theoretical and Computational Studies in:</i>	
	1.Phase Transformations in Alloys 2.Interface Chemistry and Surface Engineering 3.Defect Engineering in Alloys 4.Advanced Computational Methods Development	
	I am passionate about integrating theoretical modeling and atomistic simulations to understand and design complex alloy systems. I primarily focus on how phase transformations, defects, interfaces, and surface reactions govern the mechanical performance and environmental stability of structural alloys. By bridging atomic-scale mechanisms with mesoscale deformation and degradation behaviors, my work aims to establish physics-based principles and develop computational tools for the rational design of advanced alloys with enhanced strength, ductility, and corrosion resistance.	
Publications	Ganlin Chen , Dian Li, Yufeng Zheng, Liang Qi. Stability and growth kinetics of {112} twin embryos in β -Ti alloys. <i>Acta Materialia</i> , 263, 119520 (2024). doi:10.1016/j.actamat.2023.119520	
	Ganlin Chen , Yang Guo, Liang Qi. Computational studies of cathodic reaction activities on intermetallic particles and solute clusters in aluminum alloys. <i>Revision Returned to npj Materials Degradation</i> .	
	Ganlin Chen , Deepak V. Pillai, Yufeng Zheng, Liang Qi. Energetic origins of competing deformation modes in metastable titanium alloys. <i>Ready to submit to Materials Today</i> . arXiv:2510.13113	
	Ganlin Chen , Liang Qi. Phase-stability-dependent {332} twin growth kinetics in BCC titanium alloys. <i>In preparation</i> .	

Zhiying Liu, **Ganlin Chen**, Soumya S. Dash, Yinghao Zhou, Changjun Cheng, Mingqiang Li, Jiahui Zhang, Huicong Chen, Weibing Wang, Weicheng Xiao, Daolun Chen, Ji-Jung Kai, Liang Qi, Yu Zou. Slip heterogeneity in a colony-structured titanium alloy: planar versus wavy slip traces. *International Journal of Plasticity*, 194, 104462 (2025). [doi:10.1016/j.ijplas.2025.104462](https://doi.org/10.1016/j.ijplas.2025.104462)

Aditya Sundar, **Ganlin Chen**, Liang Qi. Substitutional adsorption of chloride at grain-boundary sites of hydroxylated alumina surfaces accelerates the initiation of localized corrosion. *npj Materials Degradation*, 5, 1–15 (2021). [doi:10.1038/s41529-021-00161-w](https://doi.org/10.1038/s41529-021-00161-w)

Chao Huang, Eva Katharine Pontrelli, Jae Wan Lee, Binyu Wang, **Ganlin Chen**, Tatiane Cogo Machado, Hemanth Maddali, Benjamin De Chazal, Anish Tuteja, Ronald G. Larson, N. Rodríguez-Hornedo, Max Shtein. *Advanced Materials*, 47, e10419 (2025). [doi:10.1002/adma.202510419](https://doi.org/10.1002/adma.202510419)

Haoyue Wangchen, **Ganlin Chen**, Le Wei, Yumo Pan, Yinghui Sun, Liang Zhao. Research progress for graphene-based anode materials of lithium-ion batteries. *Chinese Battery Industry*, 23, 269–275 (2019).

Presentations

Oral

155th TMS Annual Meeting & Exhibition, San Diego, California, 2026.

Materials Research Society Fall Meeting & Exhibit, Boston, Massachusetts, 2024.

3rd World Congress on High Entropy Alloys, Pittsburgh, Pennsylvania, 2023.

10th International Conference on Multiscale Materials Modeling, Baltimore, Maryland, 2022.

Poster

Michigan Materials Research Institute Annual Summit, Ann Arbor, Michigan, 2026

Physical Metallurgy Gordon Research Conference, Easton, Massachusetts, 2025

Physical Metallurgy Gordon Research Seminar, Easton, Massachusetts, 2025

Michigan Materials Research Institute Annual Summit, Ann Arbor, Michigan, 2024

Summer School for Integrated Computational Materials Education, Ann Arbor, Michigan, 2024